

Data, Graph, Sentence: five trio posters

AP Statistics · Unit 1 Poster · Teacher key

[BOOK] HOW THE DAY RUNS

Groups of three. Five groups, five sheets, every Unit 1 display covered: frequency table + bar chart (1), relative frequency table + relative frequency bar chart (2), dotplot (3), stem-and-leaf plot + histogram (4), parallel box plots (5). A pair takes sheet 3. Give the period's persistent misconception to a strong trio: **sheet 2 in Period B** (counts vs percents), **sheet 4 in Period E** (the mean is not resistant).

Order: the trio solves (a)–(d) together on the sheet first. Then Data Keeper, Grapher and Writer split the poster. Every poster is Data, Graph, Sentence: the data table, the required display, and the DOK-3 answer in two or three sentences with numbers from the graph.

Two days are scheduled (B: Thu 10/8 + Fri 10/9; E: Fri 10/9 + Wed 10/14). If a class finishes in one, run the gallery walk and start the Progress Check a day early.

Where the questions came from: every claim on these sheets is a mistake this class made repeatedly on Unit 1 worksheets and quizzes, per the misconception tracker (215 flagged lines, 29 students): numbers without context, numeric labels treated as quantitative, counts compared across unequal groups, the mean treated as resistant, skew direction read backwards, SOCS descriptions missing unusual features, box plot sections misread as 75%. The one Unit 1 cluster with no display behind it, z-scores read as percents, is left to the weekly DOK sheet.

Before handing out sheets: project the example poster (separate PDF) and walk its Data, Graph, Sentence boxes with the talk-track on its page 2.

[TARGET] SCORING (enter in Schoology, Posters category)

E	All five must-haves on the sheet's checklist. Display correctly drawn (scale, labels, units, key). Arithmetic right. Sentences in context with a number from the graph.	100
P	Three or four must-haves, or all five with one error of direction or arithmetic (skew direction, a fence, a percent).	85
I	Two or fewer must-haves, or the wrong kind of display for the data, or the answer agrees with the claim being judged.	60

[SPEAK] DEBRIEF PROMPTS (10 minutes after the gallery walk)

- Stand by the period's persistent poster (2 in B, 4 in E). "This one came back after we taught it. What on this poster would have stopped it the second time?"
- "Every poster's sentence has the same hidden rule. What is it?" (A number means nothing without its variable, units and the question it answers.)
- Point at sheet 4 and sheet 3 together: "Two different data sets, same lesson about the mean. Say it in one sentence."

Poster	Group	☐ ☐ ☐ ☐ ☐	E / P / I	Note
1 Freq. table + bar chart				
2 Rel. freq. table + chart				
3 Dotplot				
4 Stem-and-leaf + histogram				
5 Parallel box plots				

Superpowers: Frequency Table + Bar Chart

AP Statistics · Unit 1 Poster · Group sheet 1 of 5

[STAR] YOUR POSTER SHOWS a frequency table AND a bar chart of the same data

Data (Lesson 1.3). A random sample of 50 high school students was asked which superpower they would most like to have.

Superpower	Fly	Freeze time	Invisibility	Super strength	Telepathy
Students	9	15	7	3	16

[WARN] THE CLAIM YOU ARE JUDGING

A streaming service is pitching a show called *Freeze Time*: “Freeze time is the superpower teens want most, and it is twice as popular as flying.” A second writer says the show should be about telepathy.

DOK 1 (a) Name the individuals and the variable. Is the variable categorical or quantitative? Why does that call for a bar chart instead of a dotplot or histogram?

Key: Individuals: the 50 students. Variable: preferred superpower, categorical (labels; averaging them is meaningless). Bar charts show counts or percents of categories; dotplots and histograms need a quantitative variable on the axis.

DOK 2 (b) Check the pitch’s two claims with the table. Is freeze time the *most* chosen? Is it *twice* as popular as flying? Show the numbers.

Key: Most: no. Telepathy 16 beats freeze time 15. Twice: no. $15 \div 9 \approx 1.7$; twice would be 18.

DOK 2 (c) Write one claim about the two most popular superpowers that the data *does* support. Include a number.

Key: $16 + 15 = 31$ of 50 students, $31/50 = 62\%$, chose telepathy or freeze time. Telepathy alone is 32%, the single most popular choice.

DOK 3 (d) ★ **For the poster:** in two or three sentences, say what the data supports, what it does not, and the honest claim. Name the context and use at least two numbers.

Key: Model: “In a random sample of 50 high school students, telepathy was the most chosen superpower (16 students, 32%), just ahead of freeze time (15, 30%). Freeze time (15) was chosen more often than flying (9), but not twice as often. The honest pitch: 62% of students chose either telepathy or freeze time.”

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** the frequency table with a total row (adds to 50)
- ☐ **Graph:** bar chart: title, both axes labeled, bars separated, count scale that fits 16
- ☐ **Sentence:** individuals, variable, and “categorical” in one sentence
- ☐ **Sentence:** “most” and “twice” each judged with numbers
- ☐ **Sentence:** one honest claim, in context, with a number

All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: table, total, the “most” / “twice” arithmetic. **Grapher:** the bar chart. **Writer:** the three sentences.

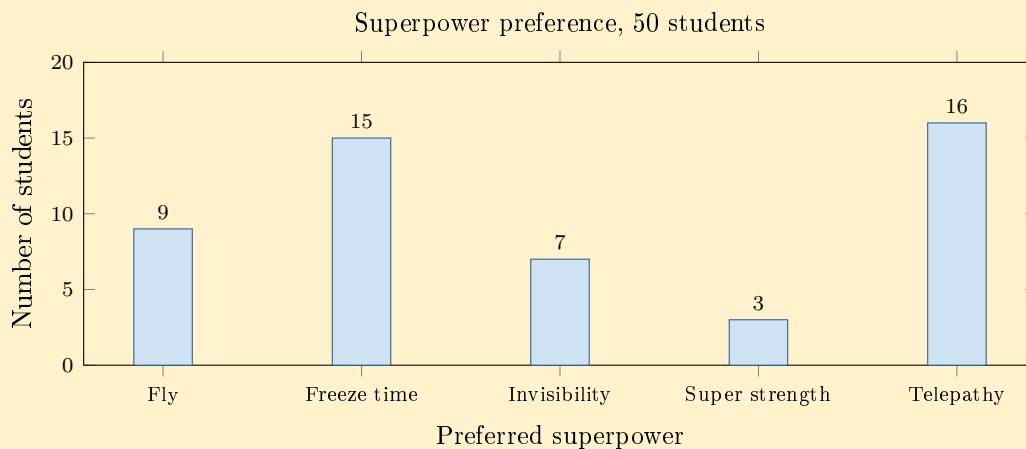
Frame (optional): In a sample of _____, the most chosen superpower was _____, with _____ students (____). Freeze time was chosen about _____ as often as flying, so the pitch is _____.

how many *who*
which *count* *percent* *how many times*
supported or not

Word bank: telepathy · freeze time · 1.7 times · twice · 32% · 30% · 62% · not supported · supported
Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

[STAR] WHAT THE FINISHED GRAPH SHOULD LOOK LIKE



Spirit Day: Relative Frequency Table + Bar Chart

AP Statistics · Unit 1 Poster · Group sheet 2 of 5

[STAR] YOUR POSTER SHOWS a relative frequency table for both grades **AND** a side-by-side relative frequency bar chart

Data (Lesson 1.4). A teacher surveyed two grades about their favorite spirit day theme.

Theme	9th grade (30 students)	10th grade (20 students)
Pajama Day	12	5
Jersey Day	6	8
Twin Day	9	3
Costume Day	3	4

[WARN] THE CLAIM YOU ARE JUDGING

Student council picked Jersey Day for the whole school: *“Jersey Day got the most 10th-grade votes, and 9th grade was close anyway: 6 versus 8.”*

DOK 1 (a) Turn every count into a percent of its own grade (divide by 30 or by 20). Check that each column totals 100%.

Key: 9th: Pajama 40%, Jersey 20%, Twin 30%, Costume 10%. 10th: Pajama 25%, Jersey 40%, Twin 15%, Costume 20%.

DOK 2 (b) Which theme does each grade actually prefer? Is “6 versus 8 is close” still true in percents?

Key: 9th prefers Pajama Day (40%); 10th prefers Jersey Day (40%). Jersey Day: 20% of 9th graders vs 40% of 10th graders, half as popular, not close. The counts only looked close because 9th grade has 30 students and 10th has 20.

DOK 2 (c) In one or two sentences: why would a chart of raw counts have misled the council, and what does dividing by the group size fix?

Key: Counts can only be compared when groups are the same size; the 30-student grade looks more enthusiastic about everything. Dividing by group size puts both grades on the same 0 to 100% scale.

DOK 3 (d) ★ **For the poster:** the grades disagree. Recommend one schoolwide theme and defend it with percents in two or three sentences. Any theme is fine if the percents back it.

Key: Example: “Pajama Day is the only theme chosen by at least a quarter of both grades (40% of 9th graders, 25% of 10th graders), so it is the fairest schoolwide pick. Jersey Day was 10th grade’s favorite at 40%, but only 20% of 9th graders chose it. The council’s ‘6 versus 8’ compared counts from groups of 30 and 20, which is why it misled them.” Jersey Day is acceptable if defended as “10th grade’s clear favorite at 40%, and 9th grade’s 20% still beats Costume Day.”

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** relative frequency table, eight percents, both columns total 100%, two divisions shown
- ☐ **Graph:** side-by-side relative frequency bar chart: title, percent axis, key for the two grades
- ☐ **Sentence:** each grade's favorite, with its percent
- ☐ **Sentence:** “6 versus 8 is close” rejected with 20% vs 40%
- ☐ **Sentence:** the recommendation defended with percents, plus why counts mislead when groups differ in size

All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: the eight percents and column totals. **Grapher:** the side-by-side chart with key. **Writer:** favorites, the recommendation, the “why counts mislead” sentence.

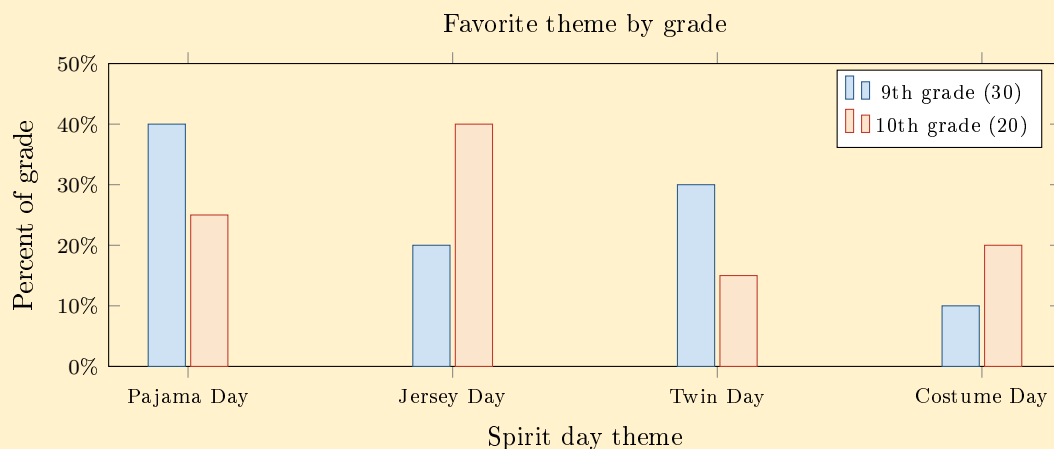
Frame (optional): Jersey Day was chosen by _____ of 9th graders but _____ of 10th graders, so the grades were _____. The council compared _____ from groups of _____ and _____, so _____.

percent percent
close or not counts or percents size size
what went wrong

Word bank: 20% · 40% · 25% · not close · close · counts · percents · 30 · 20 · the bigger grade looked more enthusiastic · the smaller grade looked more enthusiastic · Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

[STAR] WHAT THE FINISHED GRAPH SHOULD LOOK LIKE



Late Bus: Dotplot

AP Statistics · Unit 1 Poster · Group sheet 3 of 5

[STAR] YOUR POSTER SHOWS a dotplot: every value as a dot on a number line with a scale

Data (Lesson 1.6). A school recorded how many minutes 11 students waited for a late bus after school.

Student	1	2	3	4	5	6	7	8	9	10	11
Minutes waiting	1	2	3	3	4	4	4	5	5	6	18

[WARN] THE CLAIM YOU ARE JUDGING

The transportation office reports: *“Students wait an average of 5 minutes for the late bus, so there is no problem.”*

DOK 1 (a) Describe the distribution in context, one labeled line each: shape, center, variability, unusual features. (Sketch the dotplot on page 2 first if it helps.)

Key: Shape: skewed right, one peak at 4 minutes. Center: a typical wait is about 4 minutes (median 4). Variability: waits ran from 1 to 18 minutes. Unusual: a gap from 6 to 18 and one student at 18 minutes, an apparent outlier.

DOK 2 (b) Find the five-number summary and the IQR. Is 18 an outlier by the $1.5 \times \text{IQR}$ rule? Show the fence.

Key: Min 1, Q1 3, median 4, Q3 5, max 18. $\text{IQR} = 5 - 3 = 2$. Upper fence $= 5 + 1.5(2) = 8$. 18 is above 8: an outlier.

DOK 2 (c) The mean is $55 \div 11 = 5.0$ minutes and the median is 4. Which dot opens that gap? Which measure describes a typical wait, and why?

Key: The dot at 18 pulls the mean up: without it the other ten average $37/10 = 3.7$. The median stays at 4 because it depends on position only. The median describes the typical wait; the mean is not resistant to the outlier.

DOK 3 (d) ★ For the poster: explain why “average of 5, no problem” hides the real problem, then rewrite the office’s report honestly in one or two sentences, with numbers.

Key: Model: “Most of the 11 students waited about 4 minutes for the late bus, but one student waited 18 minutes, far outside everyone else’s 1 to 6 minutes. Averaging in that 18 produced the office’s ‘5 minutes’ and hid the real problem: one bus that was badly late.”

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** the 11 values, the five-number summary, the IQR
 - ☐ **Graph:** dotplot: 11 dots, number line 0 to 20 with even ticks, axis label with units, title
 - ☐ **Sentence:** shape, center, variability, unusual features, each labeled and in context
 - ☐ **Sentence:** the outlier check: Q1 3, Q3 5, IQR 2, fence $5 + 3 = 8$, so 18 is an outlier
 - ☐ **Sentence:** the honest rewrite, with mean 5.0 vs median 4 explained from the dotplot
- All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: five-number summary, IQR, fence, mean. **Grapher:** the dotplot. **Writer:** the four description lines and the honest rewrite.

Frame (optional): The distribution of _____ is _____, with a typical wait of about _____ minutes. Waits ranged from _____ to _____ minutes; _____ is an outlier because it is above the fence of _____. The office's "average of 5" misleads because _____.

variable in context shape

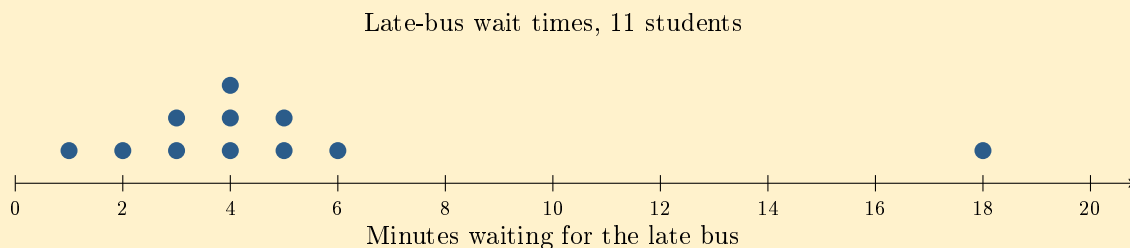
number min max which value

fence reason

Word bank: minutes waiting for the late bus · skewed right · skewed left · symmetric · 4 · 5 · 1 · 18 · 8 · one long wait pulled the mean up · the median is not resistant Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

[STAR] WHAT THE FINISHED GRAPH SHOULD LOOK LIKE



Resting Heart Rate: Stem-and-Leaf Plot + Histogram

AP Statistics · Unit 1 Poster · Group sheet 4 of 5

[STAR] YOUR POSTER SHOWS a stem-and-leaf plot AND a histogram (bins of 5 bpm) of the same 25 values, side by side

Data (Lesson 1.2 spring-sport context). Resting heart rate, in beats per minute (bpm), for 25 students who play a spring sport, already in order:

42, 46, 50, 54, 57, 58, 59, 60, 60, 61, 62, 62, 62, 63, 64, 64, 65, 65, 66, 66, 67, 68, 68, 69, 70

[WARN] THE CLAIM YOU ARE JUDGING

The coach says: *“Resting heart rate is skewed, so the mean is pulled up. I will use the mean as the typical athlete, and anyone below 50 bpm has to see the trainer.”*

DOK 1 (a) Make the stem-and-leaf plot (stems 4 to 7, leaves in order, a key such as $6 \mid 2 = 62$ bpm). Tally the bins 40–44, 45–49, 50–54, 55–59, 60–64, 65–69, 70–74. Use the sketch boxes on page 2.

Key: 4 | 26; 5 | 04789; 6 | 00122234455667889; 7 | 0. Bins: 1, 1, 2, 4, 9, 7, 1.

DOK 2 (b) Shape: which way does the tail point? Which way does that pull the mean, above or below the median? Check with the actual mean (the 25 values sum to 1528) and the median.

Key: One peak in the 60s, tail toward the LOW values: skewed left. A left tail pulls the mean DOWN. Mean = $1528/25 = 61.1$, median = 62 (13th value). The coach has the direction backwards.

DOK 2 (c) Find Q1, Q3, the IQR and the lower fence. Which athletes are outliers by the $1.5 \times \text{IQR}$ rule? Does “below 50” match?

Key: $Q1 = (58 + 59)/2 = 58.5$ (median of the lower 12). $Q3 = (66 + 66)/2 = 66$. $\text{IQR} = 7.5$. Lower fence = $58.5 - 1.5(7.5) = 47.25$. Outliers: 42 and 46. The 50 is not an outlier. The coach’s cutoff happens to catch the same two athletes, but it was not derived from the data; the fence is 47.25.

DOK 3 (d) ★ **For the poster:** in two or three sentences, which center should the coach report and why; and the trainer wants each athlete’s exact number, so which display do you hand her and what does the other one hide?

Key: Model: “Resting heart rates for the 25 athletes are skewed left with two low outliers (42 and 46 bpm), so the mean of 61.1 is pulled below the median of 62; the coach should report the median because it resists those outliers. Hand the trainer the stem-and-leaf plot: it shows every athlete’s exact value, while the histogram groups them into 5-bpm bins and hides who is at 42 and 46.”

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** the 25 values; mean 61.1, median 62, Q1 58.5, Q3 66, IQR 7.5, lower fence 47.25
- ☐ **Graph:** stem-and-leaf plot, stems 4–7, leaves in order, with a key
- ☐ **Graph:** histogram, bins 40–44 to 70–74, bars touching, axes labeled with units, title
- ☐ **Sentence:** “skewed left” with the tail direction, and mean vs median confirming it
- ☐ **Sentence:** which center to report and why; which display shows every athlete and what the histogram hides

All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: mean, median, quartiles, fence, outliers. **Grapher:** both displays, side by side. **Writer:** shape and direction, the center choice, the trainer sentence.

Frame (optional): The distribution of _____ is skewed _____, so the mean of _____ is _____ the median of _____. The coach should report the _____ because _____. The trainer gets the _____ because _____.

variable in context direction value

above or below value which reason

which display what the other hides

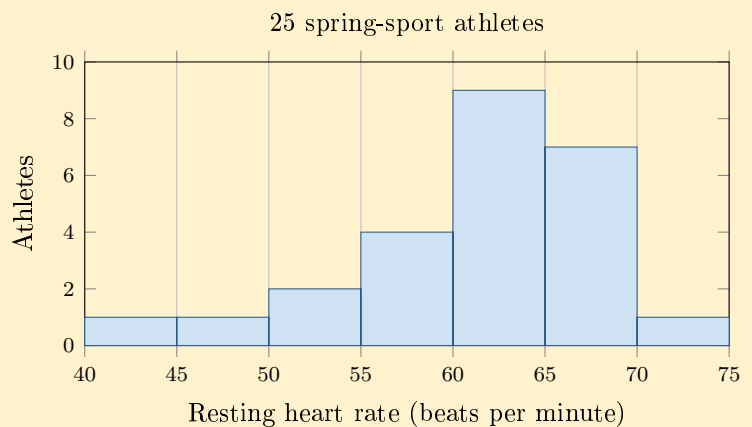
Word bank: resting heart rate in bpm · left · right · 61.1 · 62 · below · above · median · mean · it resists the two low outliers · it follows the two low outliers · stem-and-leaf plot · histogram · bins hide the exact values · Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

[STAR] WHAT THE FINISHED GRAPH SHOULD LOOK LIKE

Stem	Leaf
4	2 6
5	0 4 7 8 9
6	0 0 1 2 2 2 3 4 4 5 5 6 6 7 8 8 9
7	0

Key: 6 | 2 = 62 bpm



Study Halls: Parallel Box Plots

AP Statistics · Unit 1 Poster · Group sheet 5 of 5

[STAR] YOUR POSTER SHOWS two box plots on **ONE** shared number line, the outlier as a separate point

Data (Lesson 1.9). Nightly homework time, in minutes, for students in two study halls.

	Minimum	Q1	Median	Q3	Largest non-outlier	Outliers
Study Hall A	30	45	60	90	110	one at 150
Study Hall B	45	54	60	66	75 (maximum)	none

[WARN] THE CLAIMS YOU ARE JUDGING

The principal: *“Both study halls have a median of 60 minutes, so homework time is the same in both.”*

A parent: *“What percent of Study Hall A students spent more than 90 minutes?”*

DOK 1 (a) Sketch both box plots on the number line on page 2. Then find each study hall’s IQR and range.

Key: A: $IQR = 90 - 45 = 45$; range = $150 - 30 = 120$. B: $IQR = 66 - 54 = 12$; range = $75 - 45 = 30$.

DOK 2 (b) Compare the two: shape, center, variability, unusual features. Every line needs a comparative word (greater, less, similar, the same) and the context (homework minutes).

Key: Shape: A appears skewed right (long upper whisker, outlier); B appears roughly symmetric. Center: the median homework time is the same, 60 minutes. Variability: A’s times are far more spread out (IQR 45 vs 12; range 120 vs 30). Unusual: A has one high outlier at 150 minutes; B has none.

DOK 2 (c) Answer the parent. How does a box plot let you say that without the individual data?

Key: About 25%: Q3 is 90, and each of the four sections of a box plot holds about one quarter of the data. (Above the median at 60 is 50%, not 75%.)

DOK 3 (d) ★ **For the poster:** is the principal right? Use two features besides the median. Then: which study hall has the most predictable homework time, and which box-plot feature proves it? Two or three sentences.

Key: Model: “The principal is wrong: the medians match at 60 minutes, but Study Hall A’s times are far more spread out (IQR 45 vs 12 minutes, range 120 vs 30) and A has a 150-minute outlier while B has none. Study Hall B is more predictable: its narrower box and shorter whiskers mean the middle half of B’s students finished within a 12-minute window.”

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** both five-number summaries, both IQRs and ranges
 - ☐ **Graph:** two box plots on one scale (30 to 150), labeled A and B, axis label with units, 150 as a separate point, each section labeled 25%
 - ☐ **Sentence:** four comparison lines (shape, center, variability, unusual), each with a comparative word, in context
 - ☐ **Sentence:** the parent answered (about 25%) with the “each section is 25%” rule
 - ☐ **Sentence:** the principal judged with numbers (IQR 45 vs 12, range 120 vs 30, outlier vs none) and the predictability pick tied to the narrower box
- All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: IQRs, ranges, the 25% labels. **Grapher:** both plots on one scale, outlier as a point.
Writer: four comparison lines, the parent’s answer, the pick.

Frame (optional): The median homework time is _____ for both study halls at _____ minutes, but
comparative value
 Study Hall A’s times are _____ spread out (IQR _____ vs _____ minutes) and A has _____.
comparative value value unusual feature
 About _____ of A spent more than 90 minutes because _____.
percent box plot rule

Word bank: the same · greater · less · far more · 60 · 45 · 12 · 25% · 50% · 75% · one high outlier at 150 · no outliers · each section of a box plot holds about a quarter of the data Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

[STAR] WHAT THE FINISHED GRAPH SHOULD LOOK LIKE

